

7 Collective Action Problems Quiz

1. What does game theory study?

- A) Strategic interactions in which agents choose among actions
- B) Only single-agent perception
- C) Hardware reliability
- D) Image segmentation

ANS:

2. What is a Nash equilibrium?

- A) An outcome that maximizes total welfare
- B) A strategy profile where no player benefits by changing strategy unilaterally
- C) An outcome with equal payoffs
- D) Any cooperative agreement

ANS:

3. What is a Pareto improvement?

- A) A change that hurts everyone equally
- B) Any unilateral defection
- C) A change that helps at least one party without hurting anyone
- D) A zero-sum transfer

ANS:

4. Why is the Prisoner's Dilemma a collective-action problem?

- A) Cooperation is individually dominant
- B) The game has no equilibrium
- C) Defection never pays
- D) Individually rational defection can produce an outcome worse for both players

ANS:

5. What can repeated interaction add to a Prisoner's Dilemma?

- A) Future rewards and punishments that can support cooperation
- B) A guarantee of defection
- C) Removal of all strategies
- D) A zero-length time horizon

ANS:

6. How does Tit-for-Tat begin and respond?

- A) It always defects
- B) It cooperates first, then copies the other player's previous move
- C) It chooses randomly forever
- D) It cooperates only after two defections

ANS:

7. How can an AI corporate race resemble a Prisoner's Dilemma?

- A) Safety investment always gives the highest private short-term payoff
- B) Firms cannot affect one another
- C) Each firm may gain by cutting safety corners even though all firms are worse off if everyone does
- D) There is no competition

ANS:

8. What is a security dilemma?

- A) A system-security audit
- B) A guarantee that weapons reduce fear
- C) A failure to encrypt data
- D) One actor's defensive buildup makes others feel less secure and prompts escalation

ANS:

9. What is extortion in an iterated game?

- A) A strategy that enforces a payoff advantage over another player
- B) Any equal cooperative strategy
- C) A Pareto improvement for all
- D) A strategy with no response to others

ANS:

10. What is direct reciprocity?

- A) Helping only genetic relatives
- B) Helping someone because they may return the favor in future encounters
- C) Helping based on another person's reputation
- D) Government coercion

ANS:

11. What is indirect reciprocity?

- A) Repeated exchange between the same two agents
- B) Punishment by a central authority
- C) Helping someone because of reputation or because others may later help you
- D) Cooperation caused only by kinship

ANS:

12. What does network reciprocity rely on?

- A) All agents interacting equally with everyone
- B) No repeated contact
- C) A single global enforcer
- D) Clusters of cooperators supporting one another in a network

ANS:

13. What is kin selection?

- A) Cooperation favored because helping relatives can promote shared genes
- B) Cooperation based on public law only
- C) A market-pricing mechanism
- D) A random tournament strategy

ANS:

14. What is a public good?

- A) A private benefit with no spillovers
- B) A benefit that is non-excludable and often available to members of a group
- C) A punishment imposed on defectors
- D) A zero-sum resource only

ANS:

15. What is the free-rider problem?

- A) Everyone pays more than required
- B) Public goods are impossible to consume
- C) Individuals benefit from a collective good without contributing adequately
- D) Agents have no incentives

ANS:

16. What role can punishment play in cooperation?

- A) It always destroys cooperation
- B) It removes every cost
- C) It is effective only when costless
- D) It can deter defection, though punishment itself may be costly

ANS:

17. What is a second-order free rider?

- A)** Someone who cooperates but does not help enforce cooperation
- B)** Someone who never receives a public good
- C)** An external regulator
- D)** A player in a one-shot zero-sum game

ANS:

18. How can commitment mechanisms support cooperation?

- A)** By hiding all future actions
- B)** By making promises credible and reducing incentives to defect
- C)** By eliminating communication
- D)** By rewarding only surprise defection

ANS:

19. What can governance do for AI collective-action problems?

- A)** Guarantee that every actor is altruistic
- B)** Remove all strategic incentives
- C)** Change payoffs through standards, monitoring, agreements, or enforcement
- D)** Make coordination unnecessary

ANS:

20. Why are AI collective-action problems difficult?

- A)** All actors share identical interests
- B)** There are no externalities
- C)** Only technical capabilities matter
- D)** Private incentives, mistrust, and competition can conflict with globally safer outcomes

ANS: